SERVERLESS WEB APPLICATIONS

Building Without Managing Servers



What is Serverless?

Build and run applications without managing servers.

The cloud provider handles infrastructure, scaling and resource allocation.

Servers still exist — you just don't manage them.

How It Works

User Request
A user interacts with your app.

→

API Gateway
Routes the request to the right function.

→

Cloud Function
Executes code only when triggered.

→

Database
Stores or retrieves the required information.

→

Response
Result is returned to the user.

Why Serverless?

Auto Scaling
Adapts to changing demand.

Cost Efficient
Pay for resources used.

Less Maintenance
No direct server management.

Faster Development
Focus on application features.

Where is it Used?

E-commerce
Process orders.

Student Portals
Manage registrations.

Image Processing
Process uploaded files.

Notifications
Send event-triggered alerts.

Traditional vs Serverless

Traditional	Serverless
Manage servers	Provider-managed infrastructure
Plan capacity	Automatic scaling
Maintain infrastructure	Usage-based pricing

VS

Key Challenges

Cold Starts
Startup delays for some functions.

Vendor Lock-In
Depends on provider services.

Debugging Complexity
Harder to monitor and troubleshoot.

Popular Platforms

AWS Lambda
Amazon Web Services

Azure Functions
Microsoft Azure

Google Cloud Functions
Google Cloud

Cloudflare Workers
Cloudflare

Less server management. More innovation.

Serverless computing lets developers focus on building applications while the cloud provider manages the infrastructure.

